

Volume & Measure

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Converting Units

Every conversion is a power of ten.

Warm-Up 12 ITEMS • RECALL

1 1 m in cm

2 1 km in m

3 1 kg in g

4 2 m in cm

5 3 km in m

6 5 L in mL

7 Centimetres in a metre

8 Metres in a kilometre

9 Millimetres in a centimetre

10 Grams in a kilogram

11 Millilitres in a litre

12 Centimetres in 3 metres

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 250 cm in m

14. 1,500 g in kg

15. 0.75 km in m

16. 3,200 mL in L

17. 45 mm in cm

18. 2.5 m in cm

19. 4,500 m in km

20. 0.75 kg in g

21. 250 mL in L

22. 85 mm in cm



Counting Cubes

Volume is a count before it is a formula.

Warm-Up 12 ITEMS • RECALL

1 A 2 by 2 by 2 cube —
volume

2 A 3 by 1 by 1 box

3 A 2 by 3 by 1 box

4 A 4 by 2 by 1 box

5 A 3 by 3 by 1 box

6 A 1 by 1 by 5 box

7 Cubes in a 2 by 2 by 2
box

8 In a 3 by 1 by 1 box

9 In a 2 by 3 by 1 box

10 In a 4 by 2 by 1 box

11 In a 3 by 3 by 1 box

12 In a 1 by 1 by 5 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4 by 3 by 2 box

.....

14. A 5 by 4 by 3 box

.....

15. One layer of a 5 by 4 box holds

.....

16. That box 3 layers high

.....

17. A 6 by 5 by 2 box

.....

18. Cubes in a 3 by 3 by 3 box

.....

19. In a 4 by 3 by 2 box

.....

20. In a 5 by 4 by 2 box

.....

21. In a 6 by 2 by 3 box

.....

22. In a 5 by 5 by 2 box

.....



Working space



Error journal

Volume Formulas

Length $\times$ width $\times$ height, and a missing edge you have to find.

Warm-Up 12 ITEMS • RECALL

1 $2 \times 3 \times 4$

2 $5 \times 5 \times 2$

3 $10 \times 2 \times 3$

4 $1 \times 7 \times 4$

5 $6 \times 2 \times 2$

6 $3 \times 3 \times 3$

7 Volume of a 2 by 3 by 4 box

8 Of a 5 by 2 by 2 box

9 Of a 3 by 3 by 3 cube

10 Of a 1 by 6 by 2 box

11 Of a 4 by 4 by 1 box

12 Of a 2 by 2 by 5 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Volume 120, base area 20 — the height
.....14. Volume 100, height 5 — the base area
.....15. A 12 by 5 by 4 tank
.....16. Volume 72, edges 6 and 3 — the third edge
.....17. A cube of edge 4
.....18. Volume of a 6 by 5 by 4 box
.....19. Of a 10 by 3 by 3 box
.....20. Of a 8 by 4 by 2 box
.....21. Of a 7 by 5 by 3 box
.....22. A cube of side 4 — its volume
.....

★ Challenge NOT GRADED • REASON IT OUT

23. Two boxes: $4 \times 3 \times 2$ and $5 \times 2 \times 2$ — total volume

24. A cube of volume 125 — its edge

25. Of a 12 by 8 by 5 box

26. A cube of side 10 — its volume

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Line Plots

Fractional measurements, plotted and then reasoned about.

Warm-Up 12 ITEMS • RECALL

1 Four items at $\frac{1}{2}$ each — total

2 Eight items at $\frac{1}{4}$ each — total

3 Two items at $\frac{3}{4}$ each — total, as a/b

4 Three items at $\frac{1}{3}$ each

5 Six items at $\frac{1}{2}$ each

6 Longest of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{8}$ — type it

7 A line plot with 3 dots at $\frac{1}{2}$ — the count

8 Dots at $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$ — the total count

9 $\frac{1}{4} + \frac{1}{4}$ — type as a/b

10 $\frac{1}{2} + \frac{1}{2}$

11 Three dots at $\frac{1}{4}$ — their total as a/b

12 Two dots at $\frac{1}{2}$ — their total

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$ — total

14. Range of $\frac{1}{8}$ and $\frac{7}{8}$ — type as a/b

15. Five items totalling $\frac{5}{2}$ — the mean, as a/b

16. Difference between $\frac{7}{8}$ and $\frac{3}{8}$ — as a/b

17. Four measurements of $\frac{3}{4}$ — total

18. Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$ — the total

19. Four dots at $\frac{3}{4}$ — their total

20. The range of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — type as a/b

21. Dots at $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$ — the total

22. Six dots at $\frac{1}{3}$ — their total

Box It

Enrichment. Same volume, different shapes.

Warm-Up 4 ITEMS • RECALL

1 A box of volume 24: 2 by 3 by ?

2 A box of volume 24: 1 by 4 by ?

3 A 2 by 3 by 4 box — volume

4 A cube of side 3

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A box of volume 36: 3 by 3 by ?

6. A box of volume 60: 5 by 4 by ?

7. A 6 by 4 by 5 box

8. A box of volume 60 with base 12 — its height

Challenge NOT GRADED • REASON IT OUT

9. How many whole boxes of volume 8 fit in one of volume 96

10. A cube with the same volume as a $2 \times 4 \times 8$ box — its edge

11. Double every edge of a $2 \times 3 \times 4$ box. Volume multiplies by

12. A 12 by 5 by 4 box

13. Which holds more — type 10x2x6 or 5x4x6

14. A cube of volume 125 — its side

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Converting Units

OUT OF 22

Warm-Up • 1-12 • 100 | 1000 | 1000 | 200 | 3000 | 5000 | 100 | 1000 | 10 | 1000 | 1000 | 300

Core • 13-22 • 2.5 | 1.5 | 750 | 3.2 | 4.5 | 250 | 4.5 | 750 | 0.25 | 8.5

Challenge • 23-26 • 2500000 | 205 | 120000 | 3.5

1.2 • Counting Cubes

OUT OF 22

Warm-Up • 1-12 • 8 | 3 | 6 | 8 | 9 | 5 | 8 | 3 | 6 | 8 | 9 | 5

Core • 13-22 • 24 | 60 | 20 | 60 | 60 | 27 | 24 | 40 | 36 | 50

Challenge • 23-26 • 320 | 4 | 120 | 240

1.3 • Volume Formulas

OUT OF 22

Warm-Up • 1-12 • 24 | 50 | 60 | 28 | 24 | 27 | 24 | 20 | 27 | 12 | 16 | 20

Core • 13-22 • 6 | 20 | 240 | 4 | 64 | 120 | 90 | 64 | 105 | 64

Challenge • 23-26 • 44 | 5 | 480 | 1000

1.4 • Line Plots

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | $\frac{3}{2}$ | 1 | 3 | $\frac{1}{2}$ | 3 | 3 | $\frac{1}{2}$ | 1 | $\frac{3}{4}$ | 1

Core • 13-22 • 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 3 | $\frac{3}{2}$ | 3 | $\frac{1}{2}$ | $\frac{3}{4}$ | 2

Challenge • 23-26 • $\frac{3}{4}$ | $\frac{3}{2}$ | 1 | $\frac{1}{2}$

Fri • Box It

OUT OF 8

Warm-Up • 1-4 • 4 | 6 | 24 | 27

Core • 5-8 • 4 | 3 | 120 | 5

Challenge • 9-14 • 12 | 4 | 8 | 240 | same | 5

Count the Cubes

Build it, count it, write it. The formula comes later.

Warm-Up 12 ITEMS • RECALL

1 A 2 by 2 by 2 cube

2 A 3 by 1 by 1 box

3 A 2 by 3 by 1 box

4 A 4 by 2 by 1 box

5 A 3 by 3 by 1 box

6 A 1 by 1 by 5 box

7 Cubes in a 2 by 2 by 3 box

8 In a 4 by 1 by 1 box

9 In a 3 by 2 by 2 box

10 In a 5 by 2 by 1 box

11 In a 4 by 4 by 1 box

12 In a 1 by 1 by 8 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4 by 3 by 2 box

.....

14. A 5 by 4 by 3 box

.....

15. One layer of a 5 by 4 box

.....

16. That box 3 layers high

.....

17. A 6 by 5 by 2 box

.....

18. Cubes in a 4 by 4 by 4 box

.....

19. In a 5 by 3 by 2 box

.....

20. In a 6 by 4 by 2 box

.....

21. In a 7 by 3 by 3 box

.....

22. In a 6 by 6 by 2 box

.....

★ Challenge NOT GRADED • REASON IT OUT

23. A 10 by 8 by 4 box

24. Volume 48, base 4 by 3 — the height

25. In a 10 by 5 by 3 box

26. In a 15 by 4 by 4 box

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Layers

One layer, times the height. That is the whole idea.

Warm-Up 8 ITEMS • RECALL

1 A 3 by 4 base — cubes in one layer

2 Two such layers

3 A 5 by 2 base — one layer

4 Three such layers

5 A 6 by 1 base — one layer

6 Four such layers

7 A 2 by 5 base — one layer

8 A 4 by 4 base — one layer

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. A 7 by 3 base, 4 high

.....

10. A 8 by 5 base, 2 high

.....

11. Base area 20, height 6

.....

12. Base area 15, height 4

.....

13. Volume 100, base area 20 — the height

.....

14. A 6 by 5 base, 4 layers

.....

15. A 7 by 3 base, 5 layers

.....

16. A 8 by 4 base, 3 layers

.....

17. A volume of 90 with a 9 by 2 base — layers

.....

18. A volume of 120 with a 10 by 4 base — layers

.....



The Formula

Length $\times$ width $\times$ height, and where each one comes from.

Warm-Up 12 ITEMS • RECALL

1 $2 \times 3 \times 4$

2 $5 \times 5 \times 2$

3 $10 \times 2 \times 3$

4 $1 \times 7 \times 4$

5 $6 \times 2 \times 2$

6 $3 \times 3 \times 3$

7 Volume of a 3 by 4 by 2 box

8 Of a 5 by 5 by 2 box

9 Of a 6 by 2 by 2 box

10 Of a 4 by 3 by 3 box

11 A cube of side 2

12 Of a 7 by 2 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. A 12 by 5 by 4 tank
.....

14. A cube of edge 4
.....

15. A 9 by 3 by 2 box
.....

16. A cube of edge 5
.....

17. A 20 by 10 by 5 box
.....

18. Volume of a 9 by 4 by 3 box
.....

19. Of a 8 by 6 by 2 box
.....

20. Of a 11 by 3 by 4 box
.....

21. A cube of side 6
.....

22. Of a 10 by 10 by 2 box
.....

★ Challenge NOT GRADED • REASON IT OUT

23. A cube of volume 125 — its edge

24. A cube of volume 216 — its edge

25. Of a 15 by 8 by 6 box

26. A cube of side 12

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Find a Missing Edge

Given volume and two edges, work out the third.

Warm-Up 12 ITEMS • RECALL

1 Volume 12, edges 2 and 3 — the third

2 Volume 24, edges 2 and 3

3 Volume 30, edges 5 and 3

4 Volume 8, edges 2 and 2

5 Volume 20, edges 5 and 2

6 Volume 36, edges 6 and 3

7 Volume 24, base 12 — the height

8 Volume 30, base 10 — the height

9 Volume 36, base 9 — the height

10 Volume 20, base 4 — the height

11 Volume 48, base 24 — the height

12 Volume 18, base 6 — the height

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Volume 72, edges 6 and 3

14. Volume 120, edges 5 and 4

15. Volume 240, edges 12 and 5

16. Volume 100, height 5 — the base area

17. Volume 96, edges 8 and 4

18. Volume 120, base 20 — the height

19. Volume 144, length 6 width 4 — the height

20. Volume 210, length 7 height 5 — the width

21. Volume 96, width 4 height 3 — the length

22. A cube of volume 64 — its side

★ Challenge NOT GRADED • REASON IT OUT

23. Volume 360, edges 9 and 5

.....

24. Volume 1000 with equal edges — one edge

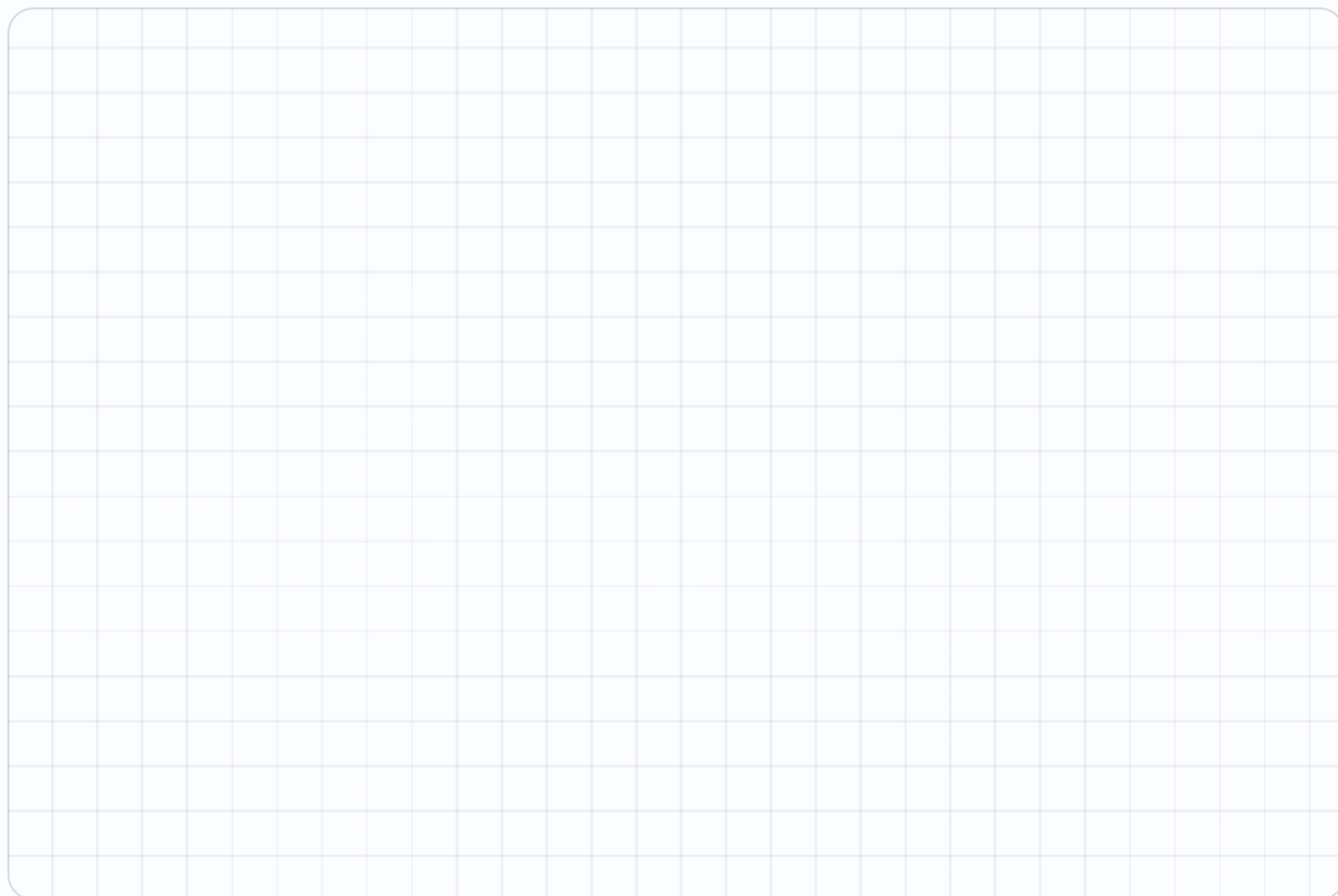
.....

25. Volume 720, length 12 width 6 — the height

.....

26. A cube of volume 343 — its side

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Cube Count

Read the drawing, call the volume before your opponent does.

NAME _____
DATE _____
RIGHT ____ OF ____ TRIED

Warm-Up 4 ITEMS • RECALL

- 1 A 2 by 2 by 3 box

- 2 A 3 by 3 by 2 box

- 3 A 2 by 2 by 2 cube — volume

- 4 A 3 by 3 by 3 cube

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A 4 by 4 by 4 cube
.....
6. A 5 by 3 by 4 box
.....
7. A 5 by 4 by 3 box
.....
8. Volume 60, base 20 — the height
.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Count the Cubes

OUT OF 22

Warm-Up • 1-12 • 8 | 3 | 6 | 8 | 9 | 5 | 12 | 4 | 12 | 10 | 16 | 8

Core • 13-22 • 24 | 60 | 20 | 60 | 60 | 64 | 30 | 48 | 63 | 72

Challenge • 23-26 • 320 | 4 | 150 | 240

2.2 • Layers

OUT OF 18

Warm-Up • 1-8 • 12 | 24 | 10 | 30 | 6 | 24 | 10 | 16

Core • 9-18 • 84 | 80 | 120 | 60 | 5 | 120 | 105 | 96 | 5 | 3

Challenge • 19-22 • 6 | 360 | 504 | 7

2.3 • The Formula

OUT OF 22

Warm-Up • 1-12 • 24 | 50 | 60 | 28 | 24 | 27 | 24 | 50 | 24 | 36 | 8 | 14

Core • 13-22 • 240 | 64 | 54 | 125 | 1000 | 108 | 96 | 132 | 216 | 200

Challenge • 23-26 • 5 | 6 | 720 | 1728

2.4 • Find a Missing Edge

OUT OF 22

Warm-Up • 1-12 • 2 | 4 | 2 | 2 | 2 | 2 | 2 | 3 | 4 | 5 | 2 | 3

Core • 13-22 • 4 | 6 | 4 | 20 | 3 | 6 | 6 | 6 | 8 | 4

Challenge • 23-26 • 8 | 10 | 10 | 7

Fri • Cube Count

OUT OF 8

Warm-Up • 1-4 • 12 | 18 | 8 | 27

Core • 5-8 • 64 | 60 | 60 | 3

Challenge • 9-14 • 44 | 8 | 216 | 240 | 6 | 64

Split the Solid

Draw the dividing line first, then measure each part.

Warm-Up 11 ITEMS • RECALL

1 A 2 by 2 by 2 plus a 2 by 2 by 1

2 A 3 by 2 by 1 plus a 3 by 2 by 1

3 A 4 by 2 by 2 plus a 2 by 2 by 2

4 A 5 by 1 by 1 plus a 3 by 1 by 1

5 Parts in an L-shaped solid

6 A 2 by 2 by 5 box

7 An L split into a 2 by 3 by 1 and a 2 by 2 by 1 — the first volume

8 The second volume

9 Their total

10 A 3 by 3 by 1 plus a 3 by 1 by 1

11 A 5 by 2 by 1 plus a 5 by 1 by 1

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. A $6 \times 4 \times 2$ plus a $3 \times 4 \times 2$

13. A $10 \times 5 \times 2$ minus a $4 \times 5 \times 2$

14. An L of $8 \times 3 \times 2$ and $4 \times 3 \times 2$

15. A $5 \times 5 \times 4$ minus a $2 \times 2 \times 4$

16. A T of $6 \times 2 \times 2$ and $2 \times 4 \times 2$

17. A 6 by 4 by 2 plus a 4 by 2 by 2

18. A 10 by 5 by 2 plus a 5 by 5 by 2

19. A 8 by 3 by 3 with a 2 by 3 by 3 removed

20. A 6 by 6 by 3 with a 2 by 2 by 3 removed

21. A 12 by 4 by 2 plus a 4 by 4 by 2

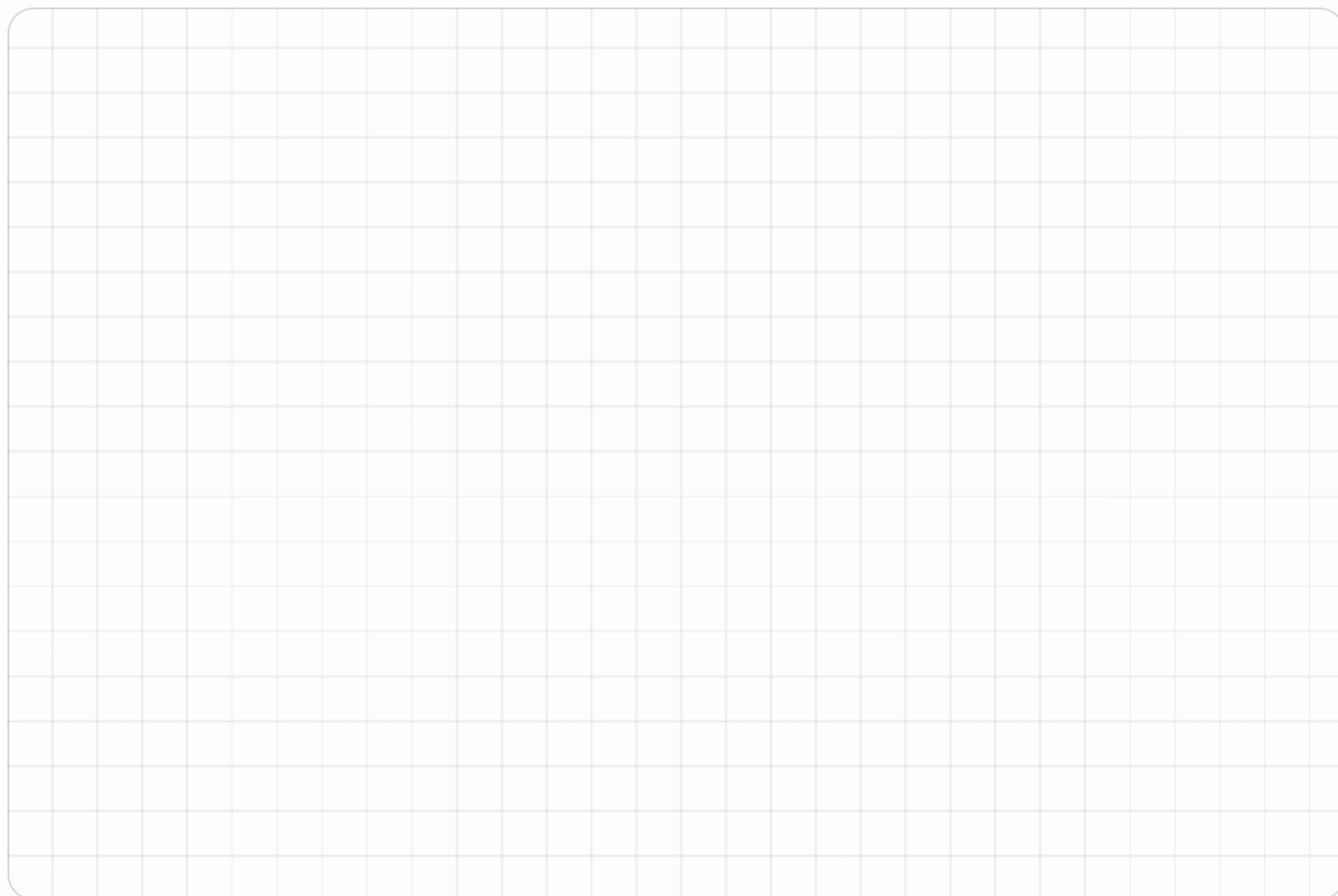
★ Challenge NOT GRADED • REASON IT OUT

22. A $10 \times 10 \times 5$ with a $4 \times 4 \times 5$ hole

23. A $12 \times 6 \times 3$ plus a $6 \times 6 \times 3$

24. A 10 by 10 by 4 with a 5 by 4 by 4 removed

25. A staircase of three 4 by 2 by 1 steps

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Add the Volumes

Two boxes, one total. Check each part before adding.

Warm-Up 9 ITEMS • RECALL

1 $12 + 8$

2 $24 + 16$

3 A $2 \times 2 \times 2$ and a $3 \times 2 \times 2$ —
total

4 A $4 \times 2 \times 1$ and a $4 \times 2 \times 1$

5 $30 + 45$

6 A $5 \times 2 \times 2$ and a $5 \times 2 \times 2$

7 $18 + 27$

8 $50 + 50$

9 $36 + 24$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. A $6 \times 5 \times 2$ and a $4 \times 3 \times 2$

11. A $8 \times 4 \times 3$ and a $2 \times 2 \times 3$

12. A $10 \times 3 \times 2$ and a $5 \times 3 \times 2$

13. Three $4 \times 3 \times 2$ boxes

14. A $7 \times 4 \times 2$ and a $3 \times 4 \times 2$

15. A 5 by 4 by 3 plus a 4 by 3 by 2

16. A 7 by 5 by 2 plus a 6 by 3 by 2

17. A 9 by 4 by 3 plus a 3 by 3 by 3

18. A 8 by 5 by 4 plus a 5 by 4 by 2

19. A 10 by 6 by 2 minus a 4 by 3 by 2

★ Challenge NOT GRADED • REASON IT OUT

20. A $12 \times 8 \times 4$ and a $6 \times 4 \times 4$

21. Five $6 \times 5 \times 2$ boxes

22. A 12 by 8 by 5 minus a 6 by 4 by 5

23. Three boxes of 4 by 3 by 2 stacked

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Line Plots

Plot measurements to the nearest eighth, then reason about them.

Warm-Up 12 ITEMS • RECALL

1 Four items at $\frac{1}{2}$ — total

2 Eight items at $\frac{1}{4}$ — total

3 Three items at $\frac{1}{3}$ — total

4 Six items at $\frac{1}{2}$ — total

5 Two items at $\frac{3}{4}$ — total

6 Longest of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{8}$

7 Dots at $\frac{1}{2}$, $\frac{1}{2}$ — the total

8 Dots at $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$

9 Three dots at $\frac{1}{2}$ — the total as a/b

10 Two dots at $\frac{3}{4}$

11 Four dots at $\frac{1}{8}$ — the total as a/b

12 Five dots at $\frac{1}{5}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$

.....

14. Range of $\frac{1}{8}$ and $\frac{7}{8}$

.....

15. Five items totalling $\frac{5}{2}$ — the mean

.....

16. Difference between $\frac{7}{8}$ and $\frac{3}{8}$

.....

17. Four measurements of $\frac{3}{4}$ — total

.....

18. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total

.....

19. Dots at $\frac{1}{8}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ — the total

.....

20. Six dots at $\frac{1}{4}$ shared among 3 — each gets as a/b

.....

21. The range of $\frac{1}{8}$ to $\frac{7}{8}$ — type as a/b

.....

22. Dots at $\frac{1}{3}$, $\frac{2}{3}$, 1 — the total

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Six pencils totalling $9/2$ inches shared equally — each

24. Two at $1/8$, three at $1/4$, one at $1/2$ — total

25. Dots at $1/2$, $1/2$, $1/4$, $3/4$ shared among 4 — each as a/b

26. Eight dots at $3/8$ — their total

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Read the Plot

Total, difference, and redistribution.

Warm-Up 11 ITEMS • RECALL

1 Five values of $\frac{1}{2}$ — total

2 Their mean

3 Four values of $\frac{1}{4}$ — total

4 Range of $\frac{1}{4}$ and $\frac{3}{4}$

5 Range of $\frac{1}{8}$ and $\frac{5}{8}$

6 The most common value on a plot is the — type mode or mean

7 Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{4}$ — the mode as a/b

8 The biggest minus the smallest is the

9 The range of $\frac{1}{4}$ and $\frac{3}{4}$ — type as a/b

10 Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$ — the mode

11 How many dots make 4 measurements

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Values $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — total

13. Values $\frac{1}{8}$, $\frac{3}{8}$, $\frac{1}{2}$ — total

14. Four values totalling 3 — the mean

15. Range of $\frac{1}{8}$ and 1

16. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total

17. That plot's mean, shared among 4 — type as a/b

18. That plot's range as a/b

19. Dots at $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$ — the total as a/b

20. Dots at $\frac{2}{3}$, $\frac{2}{3}$, $\frac{1}{3}$ — the total as a/b

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 1 m in cm

2 A 2 by 3 by 4 box

3 A 3 by 4 by 2 box —
volume

4 Centimetres in a metre

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 250 cm in m

.....

6. 1500 g in kg

.....

7. A 5 by 4 by 3 box

.....

8. Volume 120, base area 20 — the height

.....

9. Four measurements of $\frac{3}{4}$ — total

.....

10. A 6 by 5 by 4 box

.....

11. Volume 96, base 24 — the height

.....

12. 2.5 m in cm

.....

13. A 5 by 4 by 3 plus a 4 by 3 by 2

.....

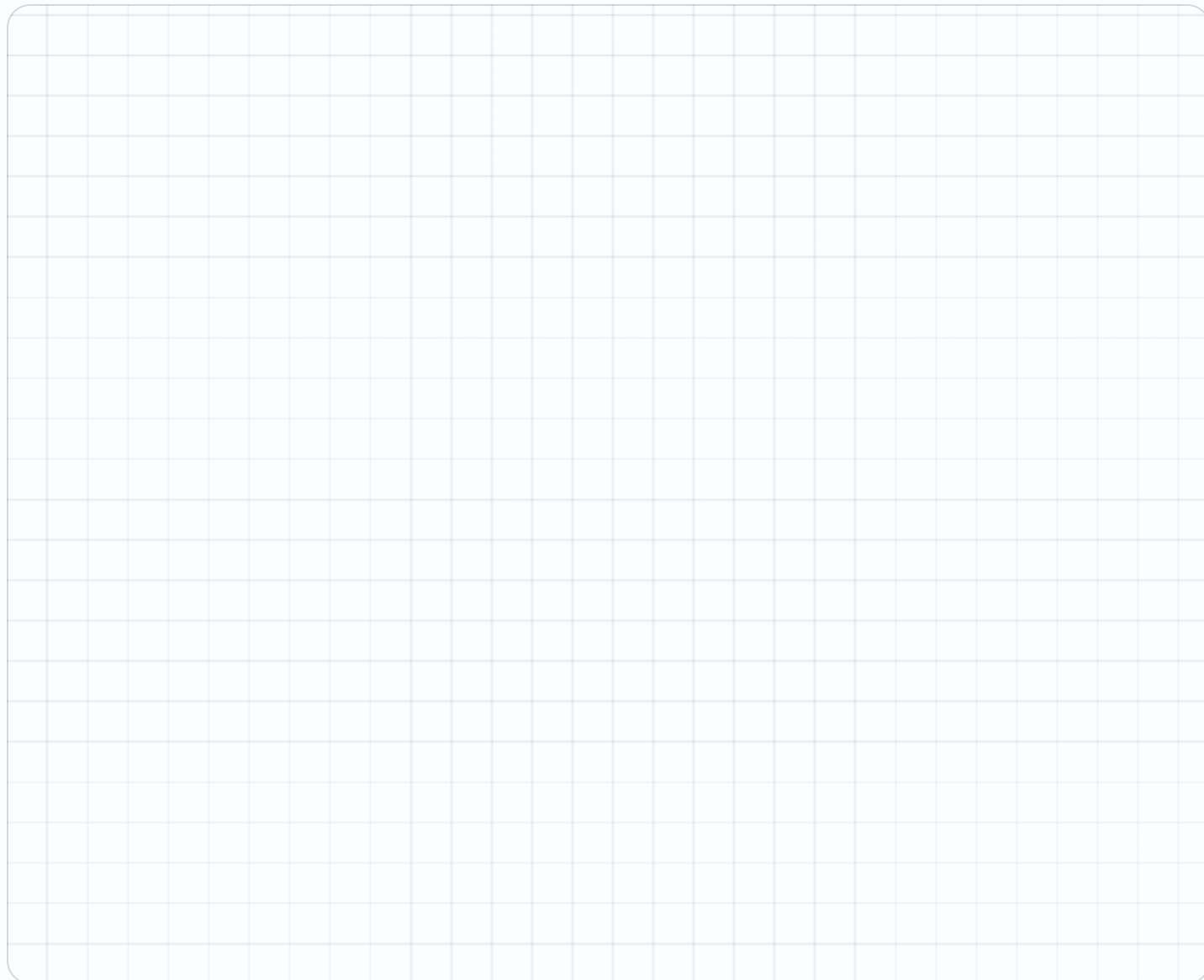
14. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total as a/b

.....

★ Challenge NOT GRADED • REASON IT OUT

15. A $10 \times 10 \times 5$ with a $4 \times 4 \times 5$ hole

16. A 10 by 10 by 4 with a 5 by 4 by 4 removed

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Split the Solid

OUT OF 21

Warm-Up • 1-11 • 12 | 12 | 24 | 8 | 2 | 20 | 6 | 4 | 10 | 12 | 15

Core • 12-21 • 72 | 60 | 72 | 84 | 40 | 64 | 150 | 54 | 96 | 128

Challenge • 22-25 • 420 | 324 | 320 | 24

3.2 • Add the Volumes

OUT OF 19

Warm-Up • 1-9 • 20 | 40 | 20 | 16 | 75 | 40 | 45 | 100 | 60

Core • 10-19 • 84 | 108 | 90 | 72 | 80 | 84 | 106 | 135 | 200 | 96

Challenge • 20-23 • 480 | 300 | 360 | 72

3.3 • Line Plots

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | 1 | 3 | $\frac{3}{2}$ | $\frac{1}{2}$ | 1 | 1 | $\frac{3}{2}$ | $\frac{3}{2}$ | $\frac{1}{2}$ | 1

Core • 13-22 • 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 3 | $\frac{3}{2}$ | 1 | $\frac{1}{2}$ | $\frac{3}{4}$ | 2

Challenge • 23-26 • $\frac{3}{4}$ | $\frac{3}{2}$ | $\frac{1}{2}$ | 3

3.4 • Read the Plot

OUT OF 20

Warm-Up • 1-11 • $\frac{5}{2}$ | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | mode | $\frac{1}{2}$ | range | $\frac{1}{2}$ | $\frac{1}{2}$ | 4

Core • 12-20 • $\frac{3}{2}$ | 1 | $\frac{3}{4}$ | $\frac{7}{8}$ | 2 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{5}{4}$ | $\frac{5}{3}$

Challenge • 21-24 • $\frac{5}{8}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 100 | 24 | 24 | 100

Core • 5-14 • 2.5 | 1.5 | 60 | 6 | 3 | 120 | 4 | 250 | 84 | $\frac{3}{2}$

Challenge • 15-16 • 420 | 320

Design Three Boxes

Same volume, different shapes. All three must check out.

Warm-Up 12 ITEMS • RECALL

1 A box of volume 24: 2 by 3 by ?

2 Volume 24: 1 by 4 by ?

3 Volume 24: 2 by 2 by ?

4 Volume 12: 2 by 3 by ?

5 Volume 36: 3 by 3 by ?

6 Volume 60: 5 by 4 by ?

7 A 2 by 3 by 4 box — volume

8 A 1 by 4 by 6 box

9 A 2 by 2 by 6 box

10 A 1 by 2 by 12 box

11 A 3 by 8 by 1 box

12 A 2 by 12 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Volume 48: 4 by 4 by ?
.....

14. Volume 72: 6 by 3 by ?
.....

15. Volume 96: 8 by 4 by ?
.....

16. Volume 120: 6 by 5 by ?
.....

17. Volume 64 with equal edges — one edge
.....

18. Three boxes all of volume 36 — one is 3 by 3 by 4, another 2 by 3 by 6, the third 1 by 6 by 6. How many
.....

19. A 36-volume box with a 9 base — its height
.....

20. A 36-volume box with a 12 base — its height
.....

21. A 48-volume box with a 4 by 4 base — its height
.....

22. A cube nearest to volume 64 — its side
.....

23. A cube with the same volume as a $2 \times 4 \times 8$ box — its edge
24. How many whole boxes of volume 8 fit in one of volume 96
25. Whole-number boxes of volume 12 with distinct dimensions — how many, counting $1 \times 1 \times 12$, $1 \times 2 \times 6$, $1 \times 3 \times 4$, $2 \times 2 \times 3$
26. A 100-volume box with a 25 base — its height

Argue for One

Which shape is best, and best for what?

Warm-Up 12 ITEMS • RECALL

1 A $2 \times 2 \times 6$ box — volume

2 A $2 \times 3 \times 4$ box — volume

3 A $1 \times 4 \times 6$ box — volume

4 Are those volumes equal — yes or no

5 A 24 cm^3 box on a 12 cm^2 base — its height

6 A cube of edge 3 — volume

7 Volume of a 2 by 5 by 5 box

8 Of a 5 by 5 by 2 box

9 Same volume — type yes or no

10 A tall thin box of 1 by 2 by 25

11 Of a 10 by 5 by 1 box

12 Of a 25 by 2 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Surface area of a $2 \times 2 \times 6$ box

.....

14. Surface area of a $2 \times 3 \times 4$ box

.....

15. Which uses less card — type $2 \times 2 \times 6$ or $2 \times 3 \times 4$

.....

16. Surface area of a cube of edge 3

.....

17. Surface area of a $1 \times 4 \times 6$ box

.....

18. Which shape wastes least material for the same volume — type cube or long thin

.....

19. A 4 by 4 by 4 box — volume

.....

20. A 1 by 8 by 8 box — volume

.....

21. A 2 by 4 by 8 box — volume

.....

22. How many of those three hold the same

.....

★ Challenge NOT GRADED • REASON IT OUT

23. For a fixed volume, the shape using least material is nearest a — type cube or slab

24. Surface area of a cube of edge 4

25. A cube of volume 216 — its side

26. A 1 by 1 by 216 box has the same volume as that cube — type yes or no

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Conversions and Volume

Both strands together, which is how they arrive in real problems.

Warm-Up 12 ITEMS • RECALL

1 1 m in cm

2 1 kg in g

3 2 m in cm

4 3 L in mL

5 A 2×2×2 box

6 500 g in kg

7 Centimetres in a metre

8 Millilitres in a litre

9 Grams in a kilogram

10 A 10 by 10 by 10 cm box
— cubic cm

11 That box in litres

12 Millimetres in a
centimetre

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 250 cm in m

.....

14. 1500 g in kg

.....

15. A 100 by 50 by 20 cm box — volume in cubic cm

.....

16. 3200 mL in L

.....

17. A 2 by 1 by 0.5 m tank — volume in cubic m

.....

18. A 20 by 10 by 10 cm box — cubic cm

.....

19. A 2 L container in cubic cm

.....

20. A 500 mL bottle in cubic cm

.....

21. A 30 by 20 by 10 cm tank in litres

.....

★

Challenge

NOT GRADED • REASON IT OUT

22. 1000 cubic cm in litres

.....

23. A 2 by 1 by 0.5 m tank in litres

.....

24. A 1 m cube in litres

.....

25. A 50 by 40 by 30 cm tank in litres

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 1 km in m

2 A 3 by 3 by 3 cube

3 1 L in mL

4 A 4 by 2 by 1 box

5 2 kg in g

6 Four items at $\frac{1}{2}$ — total

7 A 3 by 4 by 2 box — volume

8 A cube of side 3

9 Centimetres in 4 metres

10 Volume 30, base 10 — the height

11 Dots at $\frac{1}{2}$, $\frac{1}{2}$ — the total

12 2 L in mL

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 45 mm in cm

14. A 5 by 4 by 3 box

15. Volume 72, edges 6 and 3 — the third

16. 0.75 km in m

17. Range of $\frac{1}{8}$ and $\frac{7}{8}$

18. A 6 by 5 by 4 box

19. Volume 144, length 6 width 4 — the height

20. 0.75 kg in g

21. A 5 by 4 by 3 plus a 4 by 3 by 2

22. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total as a/b



Working space



Error journal

Mission 07 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 1 m in cm

2 A 2 by 3 by 4 box

3 A 2 by 3 by 5 box —
volume

4 A cube of side 4

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 250 cm in m

6. 3200 mL in L

7. A 5 by 4 by 3 box

8. A cube of edge 4

9. Volume 120, base area 20 — the height

10. Volume 96, edges 8 and 4 — the third

11. A $6 \times 5 \times 2$ and a $4 \times 3 \times 2$ — total12. Five items totalling $5/2$ — the mean

13. A 7 by 4 by 3 box

14. Volume 120, base 24 — the height

15. 3.5 m in cm

16. 4,500 g in kg

17. A 6 by 4 by 2 plus a 4 by 2 by 2

18. Dots at $1/8$, $1/4$, $3/8$ — the total as a/b

19. A cube of volume 125 — its side

20. A 20 by 10 by 10 cm box in litres

★ Challenge NOT GRADED • REASON IT OUT

21. A cube of volume 125 — its edge

.....

22. Double every edge of a $2 \times 3 \times 4$ box — volume multiplies by

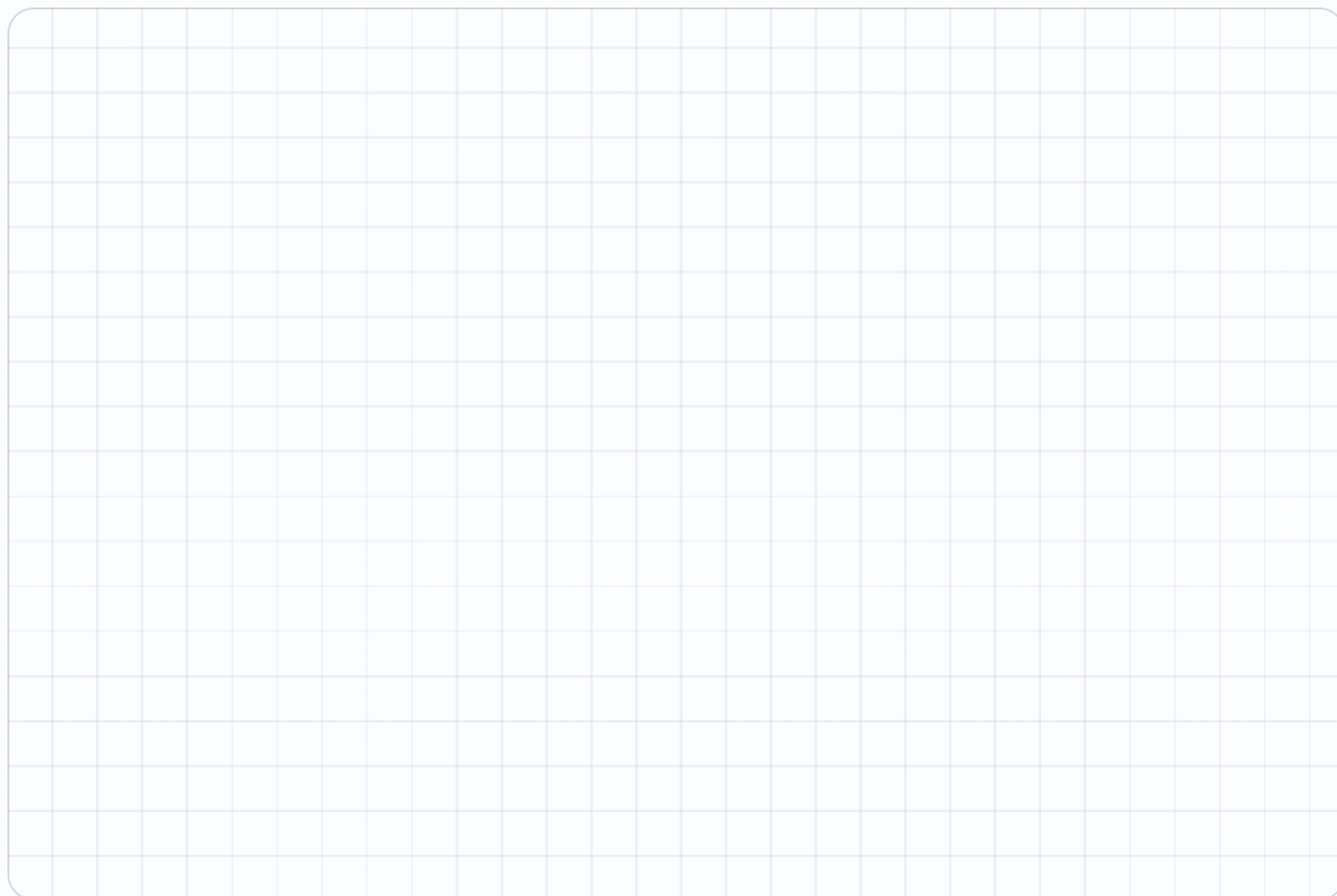
.....

23. A 12 by 8 by 5 minus a 6 by 4 by 5

.....

24. A 1 m cube in litres

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Design Three Boxes

OUT OF 22

Warm-Up • 1-12 • 4 | 6 | 6 | 2 | 4 | 3 | 24 | 24 | 24 | 24 | 24 | 24

Core • 13-22 • 3 | 4 | 3 | 4 | 4 | 3 | 4 | 3 | 3 | 4

Challenge • 23-26 • 4 | 12 | 4 | 4

4.2 • Argue for One

OUT OF 22

Warm-Up • 1-12 • 24 | 24 | 24 | yes | 2 | 27 | 50 | 50 | yes | 50 | 50 | 50

Core • 13-22 • 56 | 52 | $2 \times 3 \times 4$ | 54 | 68 | cube | 64 | 64 | 64 | 3

Challenge • 23-26 • cube | 96 | 6 | yes

4.3 • Conversions and Volume

OUT OF 21

Warm-Up • 1-12 • 100 | 1000 | 200 | 3000 | 8 | 0.5 | 100 | 1000 | 1000 | 1000 | 1 | 10

Core • 13-21 • 2.5 | 1.5 | 100000 | 3.2 | 1 | 2000 | 2000 | 500 | 6

Challenge • 22-25 • 1 | 1000 | 1000 | 60

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 1000 | 27 | 1000 | 8 | 2000 | 2 | 24 | 27 | 400 | 3 | 1 | 2000

Core • 13-22 • 4.5 | 60 | 4 | 750 | $\frac{3}{4}$ | 120 | 6 | 750 | 84 | $\frac{3}{2}$

Challenge • 23-26 • 6 | 205 | 720 | 6

Fri • Mission 07 Test

OUT OF 20

Warm-Up • 1-4 • 100 | 24 | 30 | 64

Core • 5-20 • 2.5 | 3.2 | 60 | 64 | 6 | 3 | 84 | $\frac{1}{2}$ | 84 | 5 | 350 | 4.5 | 64 | $\frac{3}{4}$ | 5 | 2

Challenge • 21-24 • 5 | 8 | 360 | 1000